

Lydia Chin

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EDUCATION

University of California, San Diego - GPA: 3.66/4

Expected June 2028

Bachelors of Science in Data Science, Double Minor in General Biology and Mathematics

La Jolla, CA

Completed Coursework: General Biology, Recombinant DNA Lab, Bioinformatics Lab, Probability Theory & Statistics

Expected Coursework: Genetics, Infectious Diseases, Biology of Cancer, Data Management, Machine Learning Algorithms

TECHNICAL SKILLS

Languages: Python, Java, SQL, MATLAB, R

Libraries: Pandas, NumPy, Matplotlib, Scikit-Learn, Seaborn, XGBoost, DESeq2, ggplot2, STRING

Tools: Git, GitHub, Microsoft Office, BLAST, NCBI/UCSC Genome Browser

Lab Techniques: PCR, gel electrophoresis, CRISPR-Cas9 genome editing, bacterial transformation, plasmid purification

EXPERIENCE

Hahcioğlu Data Science Institute

Sept 2026 - Present

Tutor

- Hold 2 office hours per week, providing 1-on-1 and small-group tutoring to undergraduates on **Python fundamentals**, **object-oriented programming**, and **data structures**

Aurion Biotech

June 2026 - Sept 2026

Manufacturing Intern (Part Time)

- Collected media lot numbers, manufacturers, and lot age across **100+** cell culture manufacturing batch records to support a reagent study for a Phase 3 cell therapy drug product
- Documented mixing parameters and timing across **150+** process steps, supporting and participating in a manager-led **risk assessment** to identify homogeneity risks throughout the manufacturing process

IDX Exchange

June 2026 - Sept 2026

Data Science Intern

- Built an **end-to-end ML pipeline** predicting California residential property closing prices from 13 months of CRMLS listing data with a train/test split by month, covering **data cleaning**, **feature engineering**, and **model evaluation**
- Trained and benchmarked four regression models (Linear Regression, Decision Tree, Random Forest, XGBoost), selecting a final XGBoost model that achieved a **0.91 R²** and **10.3% MAPE** on out-of-time test data
- Delivered a [complete repo](#) with a reproducible pipeline and worked with team to present findings and model results to company CEO and fellow interns

Boolean Lab

Oct 2025 - Present

Undergraduate Volunteer

- Processed public GEO RNA-seq datasets in Python into standardized expression, clinical, and index files for the lab's HEGEMON platform, utilizing exploratory data analysis and biological domain knowledge
- Investigating gene relationships using a boolean implication network to potentially identify cancer biomarkers

Edapt Schools

July 2025 - Sept 2025

Research Intern

- Built a two-stage LLM pipeline using **DSPy** ProgramOfThought and ReAct modules with PythonInterpreter to extract numerical data from noisy financial documents and execute multi-step mathematical computations
- Designed **YAML** test suites (basic, noisy financial, stress) and compared GPT-4-family models against GPT-5, documenting failure modes such as scattered-value extraction and interpreter library limits

TECHNICAL PROJECTS

Recipe Rating Predictor - <https://lychin24.github.io/Predicting-Recipe-Ratings/>

June 2026

- Analyzed **83K+** recipes and **730K+** user ratings from Food.com to investigate whether cooking time predicts recipe ratings, using permutation tests to establish statistical significance ($p < 0.01$)
- Built a **Decision Tree classifier** to predict whether a recipe would be rated above 4.5 stars, improving F1-score from 0.85 (baseline) to **0.94** through feature engineering and **hyperparameter tuning**

Technical Internship Predictor - <https://github.com/lychin24/Predicting-Internship-Outcomes>

Mar 2026

- Performed data cleaning, probabilistic imputation, and exploratory analysis on an undergraduate employability dataset, then evaluated regression and tree-based models via cross-validated accuracy, **RMSE**, and **R²**, determining there was no single strong predicting factor